## APPENDIX A
## ADDITIONAL ROBUSTNESS EVALUATION

We further evaluate the robustness of Spectrum-based Consistency Auditing and Recovery (SCAR) across different datasets and model architectures. For CIFAR-100 and Tiny ImageNet, we employ a ResNet-18 backbone with 100 clients, including ten compromised clients. As shown in Table I, SCAR consistently suppresses the attack across all datasets, reducing Attack Success Rate (ASR) from 96.71% to 4.09% on CIFAR-10, 99.39% to 0.34% on CIFAR-100, and 70.66% to 26.72% on Tiny ImageNet. Meanwhile, Robust Accuracy (RA) increases from 2.48% to 67.05% on CIFAR-10, from 0.31% to 39.87% on CIFAR-100, and from 7.01% to 18.10% on Tiny ImageNet, demonstrating that SCAR not only suppresses adversarial target predictions but also recovers triggered queries to their correct classes. To further evaluate the robustness of Consistency-Based Auditing (CBA), the reported detection accuracy (D-Acc.), Precision, Recall, and F1 score measure how accurately CBA determines whether an input query is triggered or benign. As shown in Table I, CBA maintains stable precision on CIFAR-100 and Tiny ImageNet, achieving 78.07% and 96.09%, respectively.

We also evaluate SCAR across different model architectures, including MobileNet and DenseNet, in addition to the default ResNet-10 backbone. As shown in Table II, SCAR consistently reduces S-ASR to less than 8% for both MobileNet and DenseNet with Benign Accuracy (BA) decreasing only about 2% to 4%. Moreover, RA improves from 2.48% to 67.05% on ResNet-10, from 18.51% to 62.12% on MobileNet, and from 22.67% to 49.99% on DenseNet. The corresponding results of CBA further show consistently high precision and recall across model architectures. These results demonstrate that the effectiveness of SCAR and the auditing capability of CBA generalize across datasets and model architectures.

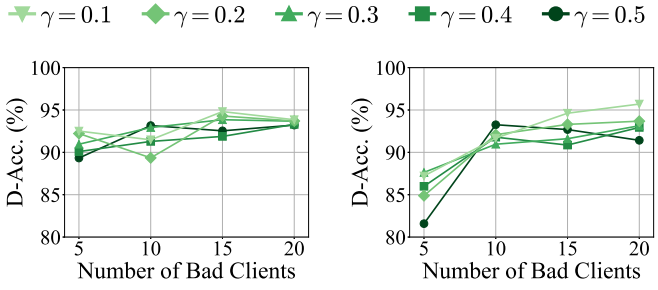
Finally, Fig. 1(a) and Fig. 1(b) show that CBA maintains a high and stable D-Acc. in different attack settings. Taken together, these results demonstrate that CBA reliably distinguishes triggered queries from benign queries across algorithms, datasets, model architectures, and varying attack configurations.

TABLE I: Performance of SCAR and CBA across different datasets. Values are reported as mean (standard deviation).

| Dataset | Defense Performance | | | | | | CBA Performance | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | U-BA | S-BA | U-ASR | S-ASR | U-RA | S-RA | D-Acc. | Precision | Recall | F1-score |
| CIFAR10 | 86.15 (5.98) | 81.06 (6.15) | 96.71 (5.80) | 4.09 (4.54) | 2.48 (3.73) | 67.05 (9.89) | 90.89 (6.52) | 92.91 (4.30) | 88.52 (12.41) | 90.18 (8.42) |
| CIFAR100 | 64.93 (8.02) | 49.91 (6.56) | 99.39 (2.27) | 0.34 (0.64) | 0.31 (0.90) | 39.87 (7.25) | 85.43 (3.35) | 78.07 (4.15) | 99.05 (1.29) | 87.25 (2.56) |
| Tiny ImageNet | 42.26 (5.82) | 41.88 (5.74) | 70.66 (19.03) | 26.72 (12.86) | 7.01 (4.83) | 18.10 (5.60) | 73.26 (10.64) | 96.09 (4.26) | 48.35 (21.58) | 61.80 (18.06) |

TABLE II: Performance of SCAR and CBA across different model architectures. Values are reported as mean (standard deviation).

| Model | Defense Performance | | | | | | CBA Performance | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | U-BA | S-BA | U-ASR | S-ASR | U-RA | S-RA | D-Acc. | Precision | Recall | F1-score |
| ResNet10 | 86.15 (5.98) | 81.06 (6.15) | 96.71 (5.80) | 4.09 (4.54) | 2.48 (3.73) | 67.05 (9.89) | 90.89 (6.52) | 92.91 (4.30) | 88.52 (12.41) | 90.18 (8.42) |
| MobileNet | 84.47 (5.65) | 80.39 (5.63) | 71.12 (23.94) | 6.42 (5.78) | 18.51 (15.11) | 62.12 (10.78) | 87.14 (5.85) | 92.31 (3.23) | 81.09 (11.99) | 85.85 (7.39) |
| DenseNet | 73.53 (8.52) | 71.93 (8.95) | 59.08 (25.74) | 7.81 (7.75) | 22.67 (15.58) | 49.99 (9.50) | 85.51 (7.27) | 95.91 (3.42) | 74.24 (14.91) | 82.82 (10.09) |




(a) FedAvg (b) FedBN

Fig. 1: CBA Acc. performance with poisoning rates from 0.1 to 0.5.